HACKATHON SUBMISSION | FINAL PRE-BUILD VERSION

# ComplianceOS

## Project Report and Build Blueprint

A vernacular, WhatsApp-first compliance and scheme co-pilot for India's small businesses. Tell them in their own language what is due, what a miss will cost, and what money they are leaving unclaimed, then get them to the next step.

| | | |
|---|---|---|
| **8.7 cr**<br>Udyam + Udyam Assist registrations, Jun 2026 [S1] | **1.67 cr**<br>active GST registrations, Jun 2026 [S7] | **31.1%**<br>MSME share of GDP, Ministry of MSME [S1] |

**How to read this report**
Every figure carries a source tag [S#] listed in Section 17. Anything marked **UNVERIFIED** had no source in our research and must not be stated as fact on a slide. Research cutoff: 30 September 2026. Rules, rates and prices change; re-check before presenting.

Prepared 1 October 2026. Informational document; not legal, tax or financial advice.

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# 1. Executive summary

**The problem.** India has a very large base of small businesses (8.7 crore registrations on Udyam and Udyam Assist as of June 2026 [S1]), over 99% of them micro [S4]. They face GST, TDS, licences, labour rules and dozens of state and central schemes spread across many portals, mostly in English and legal language. Most owners learn about an obligation when a notice or a CA's call arrives. NITI Aayog itself notes that low awareness limits how well existing MSME support works [S5].

**The product.** ComplianceOS is a WhatsApp-first co-pilot. The owner talks (Hindi voice note) or sends a photo. Five agents on one shared Business Profile return a deadline calendar, a plain-language risk number and a list of schemes they appear to qualify for, each with an official link. A Trust Layer (citations, confidence gate, consent, CA review) makes it safe to use for tax-adjacent guidance.

**The honest reframing.** Our research found that about 90% of eligible GST taxpayers already file GSTR-3B by month-end [S10]. So 'late GST return' alone is not the headline pain. The real gap is the long tail: informal micro units, non-GST obligations (licences, EPF/ESIC, labour codes), notices, scheme discovery and rule changes. We position as the *awareness and next-action layer*, not a filing tool.

## What changed from the original deck

- **5 numbers corrected or removed**: the MSME GDP figure, MSME count, '40% unawareness', '₹3.5L Cr unclaimed' and '91% match accuracy' (Section 3).
- **'GST Sahayak' is a wrong name.** The real product is 'GST Sahay' by SIDBI, an invoice-financing app [S15], not a compliance competitor.
- **Closest real competitor found**: IRIS Peridot (GSP/IRP-backed app with scheme discovery) [S16], plus VendorWise AI for document expiry tracking [S17].
- **Business model fixed**: the 15-20% CA commission conflicts with ICAI fee-sharing and solicitation rules [S18]; replaced by flat subscriptions. Ads removed.
- **PATHWAY de-risked**: guided checklist and deep links only. No credential storage, no auto-filing.
- **Free tier economics exposed**: at 12% conversion a free-heavy plan earns a thin blended margin (Section 9). Cap the free tier.

## Readiness score

| Stage | Score /100 | Why |
|---|---|---|
| Original deck | 62 | Strong problem and impact; weak feasibility, unsourced claims, no working product |
| After this report (not yet built) | 74 | Claims verified, legal map, unit economics, sharper wedge, build plan |
| Target after MVP + pilot data | 84+ | Working demo, measured accuracy, 5-8 real interviews, 1 CA pilot |

These are our own judgment scores, not a prediction of the outcome. The biggest controllable levers are a working end-to-end demo and measured numbers.

# 2. The problem, precisely

A small business owner is exposed to five compounding gaps. Fixing only one (for example, filing) leaves the rest.

| Gap | What the owner experiences | Evidence |
|---|---|---|
| Awareness | Does not know an obligation or scheme exists until a notice, fine or neighbour tells them | NITI Aayog: low awareness limits effectiveness of MSME support [S5] |
| Fragmentation | GST, MCA, Udyam, FSSAI, EPFO, ESIC, income tax/TRACES, state labour and municipal portals, each with its own login | Own count of portals; list them on the slide (not a sourced statistic) |
| Language | Portals, circulars and notices are English or legalese | Product premise; validate in interviews |
| Cost of expertise | A CA call for every doubt; pricing opaque | Own survey needed. Do not quote a CA fee range without data |
| Constant change | Rules move: GST 2.0 from 22 Sep 2025 [S8]; e-invoicing threshold lowered to ₹5 cr from Apr 2026 [S7]; Labour Codes live from 21 Nov 2025 [S11] | Regulatory records above |

## The market is big but mostly micro and informal

**8.7 cr**
registrations on Udyam + Udyam Assist, Jun 2026 [S1]

**99%+**
of Udyam registrations are micro (6.11 of 6.19 cr, Mar 2025) [S4]

**2.72 cr**
informal micro units on Udyam Assist (Jul 2025) [S3]

**1.67 cr**
active GST registrations, 30 Jun 2026 [S7]

**Implication.** Most MSMEs are not GST filers and many are informal. The paying-capable core is the GST-registered base (1.67 crore). The free, high-reach funnel is the wider Udyam base. Design the free tier for the second group and the paid tier for the first.

## What does a miss actually cost? (defensible numbers)

GST late fee is a fixed daily amount capped by turnover, plus 18% annual interest on any tax paid late [S9]. It is **not** a percentage of tax. The rates we can defend:

| Return type | Per day | Maximum per return |
|---|---|---|
| GSTR-1 / GSTR-3B, turnover up to ₹1.5 cr | ₹50 (₹25 CGST + ₹25 SGST) | ₹2,000 |
| Same, turnover ₹1.5 cr to ₹5 cr | ₹50 | ₹5,000 |
| Same, turnover above ₹5 cr | ₹50 | ₹10,000 |
| Nil return | ₹20 | ₹500 |
| Interest on tax paid late | 18% per annum | No cap |

**Penalty claim, made safe**
A monthly filer with turnover up to ₹1.5 cr who files both GSTR-1 and GSTR-3B late every month hits at most 24 x ₹2,000 = **₹48,000** a year in late fees (nil filer: 24 x ₹500 = ₹12,000), before interest and before TDS or licence penalties. So the deck's '₹5,000 to ₹50,000 avoidable penalties' is defensible only as an **illustrative worst case**. Say so on the slide, and replace it with real figures from your interviews.

**Insight that sharpens the pitch**
About 90% of eligible taxpayers file GSTR-3B by the end of the filing month (up from 68% in 2017-18) [S10]. Late GST filing is therefore a minority problem. Lead with **notices, non-GST licences, scheme discovery and rule changes**, where tools like ClearTax do not speak to the vernacular long tail.

# 3. Fact-check of the original deck

Judges check numbers. Each claim below was tested against sources. Fix every red item before submission.

| Deck claim | Verdict | What we found | Use instead |
|---|---|---|---|
| ₹16L Cr MSME contribution to GDP | **WRONG** | Ministry of MSME states MSMEs contribute **31.1% of GDP**, 35.4% of manufacturing output, 48.58% of exports (Jun 2026) [S1]. A rupee figure of ₹16L Cr has no source and looks too low for a 31% share. | '31.1% of India's GDP [S1]'. No rupee figure. |
| 6.3 Cr MSME businesses | **OUTDATED** | Udyam + Udyam Assist registrations crossed **8.7 crore** by Jun 2026 [S1]; 7.83 cr on Udyam alone by 28 Feb 2026 [S2]. 6.3 cr is an older figure. | '8.7 crore registrations, over 99% micro [S1][S4]' |
| 40% compliance failures due to unawareness | **UNVERIFIED** | No source found. NITI Aayog only says low awareness limits scheme effectiveness [S5]. | Drop it, or run a 30-person survey and quote your own result. |
| ₹3.5 Lakh Cr unclaimed schemes | **UNVERIFIED** | No source found. Related, sourced numbers: only 19% of MSME credit demand formally met, ₹80 lakh cr unmet (NITI Aayog/IFC, FY21) [S5]; Deloitte puts the gap near ₹25 lakh cr, 14% formal access [S6]. | Cite the credit-gap figures with source and year. Never call it 'unclaimed schemes'. |
| ₹5K-₹50K avoidable penalties | **PARTLY TRUE** | Consistent only as a worst case from statutory caps (up to ₹48,000/yr GST late fees alone) [S9]. Not an average. | 'Up to ₹48,000 a year in GST late fees alone, before interest'. |
| 12+ government portals | **PARTLY TRUE** | Plausible; no source. Real set: GST, MCA, Udyam, FSSAI, EPFO, ESIC, Income Tax/TRACES, DGFT (IEC), state labour, state MSME, municipal. | Show the list; call it 'our count'. |
| CA retainer ₹5K-₹20K/year | **UNVERIFIED** | No reliable source found. One blog says ₹15K-30K a year for basic GST [weak source]. | Ask 5 CAs; quote the range you hear. |
| Labour Codes 2026 rollout | **PARTLY TRUE** | Four codes in force since **21 Nov 2025**; central rules notified **8 May 2026**; state rules uneven, labour being a concurrent subject [S11][S12][S13]. | 'Labour Codes live since Nov 2025; state rules rolling out'. |
| GST Sahayak (government threat) | **WRONG** | Real name is **GST Sahay**, a SIDBI invoice-financing app built on OCEN and Account Aggregator, not a compliance tool [S15]. | Name real threats: GSTN features, Udyam/MSME portals, state portals, IRIS Peridot. |
| 91% scheme match accuracy | **INVENTED** | No test behind it. | Measure on 10 synthetic profiles against hand labels and report that. |
| 7-year encrypted vault | **PARTLY TRUE** | GST law requires records for a statutory period, but DPDP limits storage to what is needed [S19]. | Retain what law requires; give delete control; explain basis. |
| 20% women-led discount | **PARTLY TRUE** | Business choice, not a fact. Keep, but fund it in the unit economics. | Show cost in Section 9. |
| SENTINEL alert: deadline 20th to 15th | **PARTLY TRUE** | Illustrative only. GSTR-3B monthly due date is the 20th; quarterly filers 22nd/24th [S9]. | Label 'simulated alert' on the slide. |

# 4. Market size and beachhead

We size bottom-up from the GST base because that is where paid conversion is realistic. Assumptions are explicit so judges can challenge them.

| Layer | Definition | Size | Basis |
|---|---|---|---|
| Reach pool | Udyam + Udyam Assist registrations | 8.7 crore | Ministry of MSME, Jun 2026 [S1] |
| TAM (value) | All 1.67 cr active GST registrations x ₹299 x 12 | ₹5,996 crore / year | GSTN count [S7] x our price. Theoretical ceiling |
| SAM | 25% of GST base in Hindi-speaking states x 70% small taxpayers | 29.2 lakh users, ₹1,049 crore / year | **ASSUMPTIONS**: replace with GSTN state-wise data |
| SOM (3 yrs) | 0.5% of SAM | 14,623 users, ₹5.2 crore / year | **ASSUMPTION**: realistic for a bootstrapped team |

Get the real state-wise active GST payer counts from the Department of Revenue dataset (listed on Dataful) and replace the 25% and 70% guesses. That one edit turns an assumption into a sourced number.

## Why Uttar Pradesh is the right beachhead

- **Momentum.** UP led new GST registrations in H1 2026 (nearly 8 lakh new registrations nationally; total 1.67 cr on 30 Jun 2026) [S7].
- **Scheme-rich.** UP runs One District One Product (ODOP, all 75 districts), CM YUVA (loans up to ₹5 lakh project cost at 0% interest for eligible youth), and the UP MSME Promotion Policy 2022 with capital subsidy, interest subsidy, stamp duty relief and EPF reimbursement [S23].
- **Hindi-first.** One language covers the pilot. Add regional languages after the vertical slice works.
- **Field access.** Pick one district cluster your team can visit in person. Pilots need physical trust, not ads.

## First customer segments

| Segment | Why | Entry channel |
|---|---|---|
| GST-registered traders and small manufacturers (turnover under ₹1.5 cr) | Have deadlines, notices and ODOP or credit-linked scheme potential | Local CAs, trade associations |
| Women-led micro units on Udyam | Many scheme-eligible; low awareness; social-impact story | Self-help groups, bank MSME officers, CSCs |
| Small CA and tax-practitioner firms | Want a client dashboard and fewer repetitive questions | Direct outreach; flat subscription |

# 5. Legal, regulatory and safety map

This is the section technical and business judges probe hardest. Each row says what the rule is, why it matters, and how the design already responds.

| Area | What the rule says | Risk to us | Design response |
|---|---|---|---|
| GST late fee and interest | Late fee under s.47: ₹50/day (₹20 nil) with turnover caps; interest 18% p.a. under s.50 [S9] | Wrong calculation = wrong advice | Deterministic calculator in code; LLM only explains; cite section |
| Portal automation | GST and other portals use OTP and credential flows. Whether a third party may log in and file on a user's behalf depends on GSP/ASP arrangements. **Not verified in our research.** | Liability, ToS breach, account lockout | No credential storage, no auto-filing. Deep links and checklists only. Verify GSP/ASP position with GSTN or a lawyer before any roadmap claim |
| ICAI professional rules | Clause 6 (Part I, First Schedule) bars solicitation; clause 7 bars advertising; fee sharing with non-members is restricted [S18]. ICAI's 13th Edition (from 1 Apr 2026) eased advertising format, but testimonials, fee disclosure and comparative claims remain barred [S18] | 15-20% commission on CA consultations may be treated as fee sharing; star ratings may look like testimonials | Flat monthly subscription for CAs, no per-consultation cut. Verification badge and response-time stats instead of star reviews, pending a CA's written opinion |
| DPDP Act and Rules 2025 | Rules notified 13/14 Nov 2025. Consent-manager framework from Nov 2026. Full obligations from **13 May 2027**. Penalties up to ₹250 crore [S19] | We hold PAN, GSTIN, payroll and notice scans | Granular consent, purpose limitation, data minimisation, user delete, breach plan, India-region hosting, no sale of data |
| Labour Codes | In force from 21 Nov 2025; central rules 8 May 2026; state rules uneven [S11][S13][S14] | Stale or wrong state status | SENTINEL stores state status with a 'last verified' date; show it to the user |
| WhatsApp platform | Template categories, opt-in required; marketing messages cost about 7.5x utility [S20] | Account bans, cost blowout | Opt-in on first message; reminders as utility templates; no cold marketing |
| Lending and referrals | Loan referral/DSA and bank referral need regulated partners. **Not researched in depth.** | RBI digital-lending exposure | Out of MVP and out of pitch core. One roadmap line at most |
| AI advice | Tax-adjacent guidance from an LLM | Hallucinated deadline | Citations, confidence gate, CA review, visible disclaimer 'informational, not professional advice' |

**Highest-value legal step**

Before the final round, get one 30-minute call with a practising CA and, if possible, a lawyer. Ask three things: (1) is a flat CA subscription clean under ICAI rules, (2) is a deep-link checklist clearly outside GSP/ASP territory, (3) what disclaimer wording do they want. Quote their answers on the trust slide.

# 6. Competition and positioning

| Player | What they do (verified) | Gap we exploit | Threat |
|---|---|---|---|
| ClearTax | GST automation, AI-assisted reconciliation, e-invoicing [S16] | Built for users who already know what to file; English-first | High if they add vernacular voice |
| IRIS Peridot | App by IRIS (a GSP and IRP): GST number checks, filing-status checks, **scheme discovery for MSMEs**, notifications; MoUs with Telangana, Goa, Karnataka [S16] | No WhatsApp-first voice flow, no document-to-calendar, no CA trust layer evident from its listing | **Highest overlap on schemes.** Must beat it on voice, documents and trust |
| VendorWise AI | Upload GST, PAN, Udyam, ISO, FSSAI, EPF/ESIC docs; AI classifies and tracks expiry; routes to licensed professionals [S17] | Vendor-compliance angle; no scheme matching or regulatory feed visible | Medium |
| Virtual accounting services | ClearTax Business, EbizFiling, CorpBiz and similar outsource bookkeeping and filing to CAs | Paid, human, English; not an awareness layer | Low to medium |
| GST Sahay (SIDBI) | Invoice-based lending app using GSTN, Account Aggregator, OCEN [S15] | Credit product, not compliance | Adjacent; possible partner |
| Government portals | Udyam, MSME Champions grievance portal, state MSME portals (for example msme.up.gov.in) | Each is single-purpose; none unifies compliance plus schemes in a local language | Structural: government may add features |
| Tally, Zoho, Vyapar and others | Accounting/billing suites. **Not researched here.** | Verify their compliance-reminder features before claiming a gap | Unknown; check |

## Positioning statement

*For Hindi-speaking small business owners who do not know what they owe or what they could claim, ComplianceOS is a WhatsApp co-pilot that listens, reads their documents, and tells them the next action with an official source. Unlike filing tools it starts from awareness, and unlike scheme portals it ties schemes to the owner's actual deadlines and documents.*

## Defensibility, stated honestly

- **Vernacular voice access** is a real wedge today, but others can copy it. Moat comes from the compounding assets below.
- **Verified scheme and rules database** with citations: slow to build, high trust once it exists.
- **Document vault and calendar** create switching cost only after users upload real documents. Hence the DPDP-safe design matters.
- **CA network** grows in value with scale, but is thin at the start (cold-start risk). Start with 3 to 5 friendly CAs, not a marketplace.
- Do not claim 'irreversible switching costs' or 'government trust' in the pitch; neither exists yet.

# 7. Final product definition

**One sentence:** send a voice note or a photo on WhatsApp in Hindi; get back what is due, what a miss costs, and what you may qualify for, each with an official source.

## Three users, one flow

| User | Situation | What they get |
|---|---|---|
| Ramesh, trader (GST, turnover ₹80 lakh) | Gets a notice in English; calls a CA for every doubt | Photo of notice → plain-Hindi summary, due date on calendar, penalty at risk, 'send to my CA' |
| Sunita, women-led tailoring unit (Udyam, no GST) | Never heard of subsidy schemes | 5-minute voice chat → profile → 3 to 5 scheme matches with 'why you match' and required documents |
| CA Mehta, 60 small clients | Chases clients for documents; answers the same questions | Read-only client dashboard: calendar, documents, pending items; approve or flag before action |

## The five agents as one pipeline

| Agent | Input | Core logic | Output | Quality metric (measure it) |
|---|---|---|---|---|
| ARIA intake | Hindi voice note or text | Speech-to-text → LLM slot-filling into a fixed profile JSON; asks one missing field at a time and explains why | Business Profile | Profile completeness after 5 turns; STT word-error on 20 test clips |
| VEDA documents | Photo or PDF | Vision LLM extracts fields with evidence quotes; code validates GSTIN checksum, PAN format, date sanity; low confidence → ask user | Calendar items, Health Score | Field-level exact match on 30 documents; recall of low-confidence flags |
| SCOUT schemes | Profile + extracted facts | **Deterministic rules engine** over a hand-verified scheme DB; LLM only explains in Hindi | Ranked schemes: 'X of Y criteria met', missing info, official link | Precision and recall vs hand labels on 10 synthetic profiles |
| SENTINEL rule changes | Official notification pages | Ingest → dedupe → LLM tags affected attributes (state, sector, turnover band) → match profiles → plain-language alert | Targeted alert + changelog | Relevance precision on 10 real notices; false-alert rate |
| PATHWAY action | Any due item or scheme | Template checklist: documents, official deep link, 'share with my CA'. **No login, no filing** | Step list + link | Task completion in user tests |

## Trust Layer (applies to every agent, and is the differentiator)

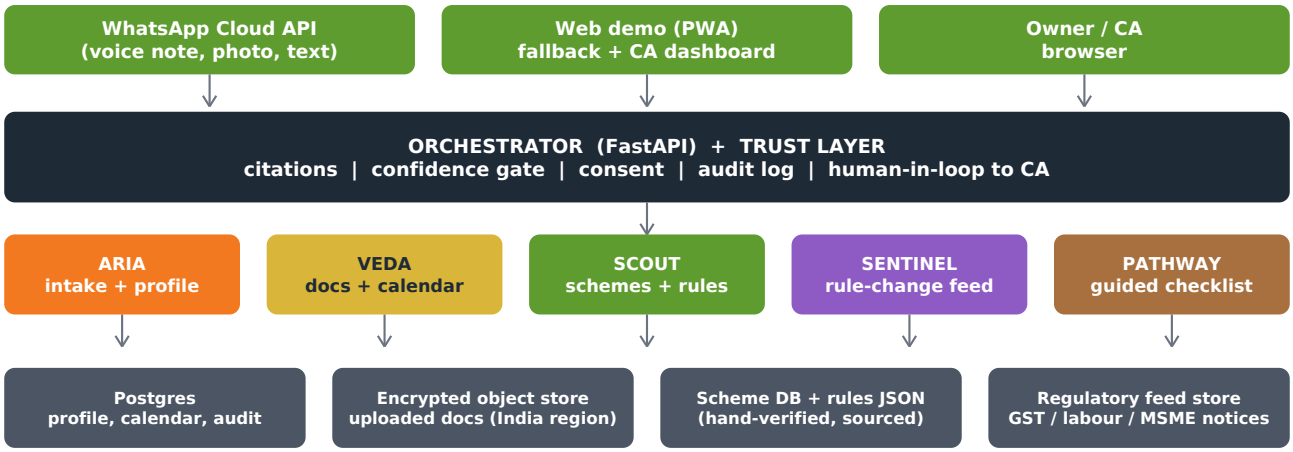
- **Citations**: every deadline, fee and scheme claim shows the source URL and 'last verified' date.
- **Confidence gate**: below 0.80, or any failed validation, the item is shown as 'please confirm' and never saved silently.
- **Deterministic where it matters**: penalties and eligibility come from code and data, not from the LLM's memory.
- **Human in the loop**: high-stakes items (notice replies, scheme applications) are shareable to a CA before the user acts.
- **Consent, control, disclaimer**: opt-in on first message, plain-Hindi privacy note, one-tap delete (DPDP-aligned [S19]); 'informational, not professional advice'; never ask for password, OTP or Aadhaar.

## Compliance Health Score and the headline number

Simple, explainable, tunable. 'Scheme headroom' sums **maximum** caps, labelled 'up to'; it is not a promised payout.

```
score = max(0, 100 - sum(penalties))
  overdue critical item (GST return, licence expired)   -15 each (max -45)
  overdue standard item                                  -8 each (max -24)
  due within 7 days                                      -4 each (max -12)
  mandatory registration missing (e.g. GST above limit)  -10
  document needed but not uploaded                       -2 each (max -6)
penalty_at_risk_inr = sum( gst_late_fee(...) + interest ) over open items     # deterministic, cited
scheme_headroom_inr = sum( max_benefit_inr ) over schemes where eligible or possible   # label "up to"
```

# 8. Architecture and technology choices




*External: Speech-to-text (Sarvam / Bhashini) | LLM + vision (Claude API) | official sources (gst.gov.in, PIB, Gazette, labour ministry)*

| Layer | Recommended choice | Why | Verified facts and cautions |
|---|---|---|---|
| Channel | WhatsApp Cloud API; web PWA fallback | Where owners already are; voice notes and photos built in | Template messages are billed per message; utility ₹0.115, marketing ₹0.8631, plus 18% GST [S20]. Use the web fallback if number approval slips |
| Speech | Sarvam speech-to-text (22 languages, Hinglish code-mixing); Bhashini as a backup | Hindi-English switching is the normal case for real users | Sarvam lists ₹1.5 per minute with free signup credits [S21] and a startup program with API credits [S21]. Bhashini offers speech, translation and text-to-speech APIs, free for non-commercial and discounted for commercial use [S22]. Benchmark both on your own clips |
| LLM + vision | Claude API (model string claude-sonnet-5-5) with JSON output and schema validation | Strong structured extraction from photos; reasoning for plain-language explanation | Measure cost per document in your own tests; replace the ₹12 assumption in Section 9 |
| Backend and data | Python FastAPI + cron/Celery; Postgres; encrypted object storage in an India region | Fast to build; DPDP-aligned | Thin orchestrator so every agent call is traceable; log consent and deletion; never log raw PAN or images |
| Rules engine | JSON rules + a small Python evaluator (Section 12) | Auditable eligibility, no LLM guessing | Each scheme needs source URL, retrieved date and checker name |
| Regulatory feed | Scheduled fetch of 2 to 3 official notification pages; hash to detect change | Real data beats a mock | Respect site terms and rate limits; keep raw text and URL for citation |

## Document pipeline (VEDA) in five steps

- Receive image or PDF → virus-scan, size-limit, store encrypted; rotate and de-skew, keep original.
- Vision LLM returns JSON with *evidence quotes* and per-field confidence (prompt in Section 12).
- Code validation: GSTIN checksum, PAN regex, date and amount plausibility, cross-field consistency (state code vs address).
- Confidence gate: any miss → 'please confirm' card in Hindi. User taps to accept or edit.
- Only confirmed fields write to the calendar; every write is logged with source document and time.

## What is real in the demo and what is simulated

| Component | Status in demo | Say this out loud |
|---|---|---|
| Hindi voice → profile | Real | Live speech, real model |
| Document extraction | Real on sample docs | Reports measured accuracy on 30 documents |
| Scheme matching | Real over 30 to 50 verified schemes | Rules engine, every card has a source |

| Component | Status in demo | Say this out loud |
|---|---|---|
| SENTINEL feed | Half real: 2 live pages ingested; one circular injected | 'This alert is simulated to show targeting' |
| PATHWAY | Real checklists and deep links | 'We guide, we never log in or file' |
| CA dashboard | Read-only prototype | 'Prototype; one CA pilot planned' |
| Payments | Simulated | 'Not live' |

# 9. Unit economics

The aim is a one-slide answer to 'can this pay for itself?'. Facts come from rate cards; items marked ASSUMPTION are placeholders to replace with your measured costs.

| Monthly cost per paying user (₹299 plan) | ₹ |
|---|---|
| WhatsApp utility templates (30 alerts/reminders) | 4.07 |
| WhatsApp marketing templates (3, opt-in digests) | 3.06 |
| WhatsApp service replies (100, worst case after 1 Oct 2026) | 13.57 |
| Speech-to-text (6 min) | 10.62 |
| Document extraction, LLM + vision (ASSUMPTION) | 12.00 |
| Chat LLM for ARIA / explanations (ASSUMPTION) | 10.00 |
| Hosting, storage, monitoring (ASSUMPTION) | 4.00 |
| Payment gateway 2% of ₹299 (ASSUMPTION) | 5.98 |
| **Total variable cost** | **63.30** |
| **Gross margin on a paying user** | **79%** |

WhatsApp rates: India utility and authentication ₹0.115, marketing ₹0.8631, plus 18% GST; service messages are free inside the 24-hour window with a paid per-message charge reported from 1 Oct 2026 after 1,000 free per business number [S20]. Sources differ on the service-message detail, so we use the conservative case and flag it for re-check on Meta's rate card.

## The free tier is the hidden cost

A free user who chats, sends voice notes and uploads documents costs money too. Uncapped, we estimate about ₹41 a month; capped (1 document, short voice, weekly digest) about ₹16.

| Free-to-paid conversion | Free uncapped (₹41) | Free capped (₹16) |
|---|---|---|
| 5% | -180% blended margin | -20% blended margin |
| 8% | -78% blended margin | 19% blended margin |
| 12% | -21% blended margin | 41% blended margin |
| 20% | 24% blended margin | 58% blended margin |

**Design rule from the numbers**
Cap the free tier hard: 1 document a month, 3 scheme matches, text and short voice only, weekly digest instead of daily reminders. Treat free as a funnel with a budget, not a gift. With a capped free tier, margins stay healthy even at 5 to 8% conversion.

## Illustrative month-12 picture (assumptions, not a forecast)

| Item | Value |
|---|---|
| Registered users | 2,000 |
| Paying (12%) | 240 |
| CA partners at ₹999 | 25 |
| Monthly recurring revenue | ₹96,735 |
| Variable cost (capped free tier) | ₹42,565 |
| Contribution before fixed costs | ₹54,170 |

| Item | Value |
|---|---|
| Payback on ₹400 CAC (ASSUMPTION, partner-led) | 1.7 months |

Fixed costs (team, tools, compliance review) are deliberately excluded because they depend on your team. Add them before you show a break-even slide.

# 10. Business model (simplified and legally safer)

Judges reward focus. Three revenue lines, single prices, no ads, no commissions on professional fees.

| Plan | Price | Includes | Notes |
|---|---|---|---|
| Free (funnel) | ₹0 | Voice onboarding, profile, Health Score, calendar of up to 5 items, 3 scheme matches, 1 document a month, weekly digest | No ads: ads on a compliance product weaken trust |
| Growth | ₹299 / month | Fair-use document uploads, SENTINEL targeted alerts, full scheme matching with checklists, 7-3-1 day reminders, CA share link | Women-led enterprises: 20% off (₹239). Fund it from partner sponsorship later |
| CA Partner | ₹999 / month (up to 25 clients); ₹2,499 (up to 100) | Client dashboard, calendar and document view, comments, approve or flag | Flat fee only. No cut of professional fees (ICAI risk [S18]) |

## Removed from the original model, and why

| Removed | Reason |
|---|---|
| 15-20% commission on CA consultations | Conflicts with ICAI fee-sharing and solicitation rules [S18]; replaced by flat CA subscription |
| Ads and sponsored slots on free tier | Trust. Revisit only for clearly separate, non-targeted partner content |
| Loan DSA, bank referral fees, FINPILOT | Needs regulated partners; scope creep; not researched in depth. One roadmap line |
| Pro tier (₹999 to ₹1,499) with API access | Two tiers are easier to explain; add later based on demand |
| Nepal and Bangladesh, B2G data licensing | Distraction at this stage; keep as long-term vision only |

## Later revenue options (roadmap, not pitch core)

- Trade-association and chamber bulk plans.
- Bank and NBFC partnerships for scheme-led customer education (subject to regulatory advice).
- State MSME department pilots for awareness delivery (grant or licence).

# 11. Validation and go-to-market

## Week 0 to 2: validate before you over-build

Goal: 8 owner interviews and 3 CA interviews. Output: one real quote for slide 1, three real numbers for slide 2, and a yes or no on willingness to pay.

### Owner interview script (15 minutes)

- Tell me about the last time you got a notice, a fine, or missed a deadline. What happened, what did it cost?
- Who do you call first? What do they charge? How long did it take to resolve?
- Which of these do you track today: GST return, TDS, licence renewal, EPF/ESIC, Udyam? How?
- Have you ever applied for a government scheme or subsidy? How did you hear about it? What stopped you?
- Do you use WhatsApp daily? Do you send voice notes? Would you send a photo of a notice to a number?
- Show the demo flow on paper. What is confusing? What would you not trust?
- Would you pay ₹299 a month for this? What would make it a clear yes?
- Who else in your market should I speak to?

### CA interview script (15 minutes)

- Which client questions repeat most? Which documents do you chase?
- Would a client dashboard save you time? What would you pay per month?
- What would worry you about an AI giving guidance to your clients?
- Is a flat monthly subscription for a dashboard acceptable to you professionally? (ICAI check)

## Week 3 to 8: pilot

- 25 businesses through 2 to 3 friendly CAs and one trade association in one district cluster.
- Track: onboarding completion, documents uploaded, calendar items confirmed, schemes viewed, return visits at day 7 and day 30, willingness to pay.
- Collect 3 short written testimonials from owners (not CAs, to avoid ICAI issues).

## Channels, in order of expected efficiency

| Channel | Why | Cost |
|---|---|---|
| Local CAs and tax practitioners | They already hold the client relationship and gain from fewer repetitive queries | Free dashboard during pilot |
| Trade associations and chambers | One meeting reaches hundreds of owners; trust comes with the endorsement | Time |
| District Industries Centres, bank MSME officers | Natural point to talk about Udyam and schemes | Time |
| Common Service Centres and local digital kiosks | Assisted onboarding for low-literacy users | Revenue share later |
| WhatsApp referral loops | Owners share with peers; free tier fuels it | Capped free cost (Section 9) |

## Milestones

| When | Milestone | Proof |
|---|---|---|
| Hackathon | Working vertical slice, measured accuracy, 5 to 8 interviews | Demo + data slide |
| +30 days | 25-business pilot live | Retention, quotes |
| +90 days | 3 CA partners paying; 300 registered users | MRR, funnel |
| +6 months | Second language; SENTINEL with 5 sources; first trade-body deal | Contract, uptime |
| +12 months | 2,000 users; 240 paying; consent-manager ready (DPDP phases [S19]) | Cohorts |

# 12. MVP build guide

Build **one vertical slice** that runs end to end, then polish it. Five half-working agents lose to one flow that works every time.

## Scope

| Priority | Feature | Real or simulated | Done when |
|---|---|---|---|
| P0 | Hindi voice or text → Business Profile (ARIA) | Real | Profile fills in under 5 turns on 5 test speakers |
| P0 | Photo of GST certificate or notice → confirmed fields → calendar (VEDA) | Real | Works on 30 sample docs; low-confidence fields flagged |
| P0 | Health Score + penalty-at-risk + scheme headroom card | Real (formula in Section 7) | Numbers reproducible from rules |
| P0 | 3 to 5 scheme matches with 'X of Y met' and official link (SCOUT) | Real | 30 to 50 verified schemes in DB |
| P0 | Trust Layer: citations, confidence gate, consent, disclaimer | Real | Visible in the demo |
| P1 | SENTINEL: 2 live pages ingested + 1 injected circular → targeted alert | Half real | Alert reaches the right profile only |
| P1 | PATHWAY checklist with deep links | Real, light | 3 checklists (GST return, Udyam, one scheme) |
| P1 | Agent trace panel | Real | Shows ARIA → VEDA → SCOUT → SENTINEL calls |
| P2 | CA read-only dashboard | Prototype | One page with a client's calendar and documents |
| P2 | Second language | Real if time | One more language through the same STT |
| Cut | Payments, auto-filing, loans, ads, API access | Do not build | n/a |

## 14-day plan (adjust to your real calendar)

| Days | Focus | Outputs |
|---|---|---|
| 1-2 | Decide stack; set up repo, DB, WhatsApp test number or web chat; start interviews | Hello-world agent; 3 interviews booked |
| 3-4 | ARIA: voice → STT → profile JSON; Hindi prompts; test with 5 voices | Profile working |
| 5-7 | VEDA: extraction prompt, validators, confirm card, calendar; collect and label 30 docs | Extraction on sample docs |
| 5-7 (parallel) | Data: build scheme DB (30 to 50), penalty rules; each with source, date, checker | Verified scheme DB |
| 8-9 | SCOUT rules engine + explanation; Health Score; PATHWAY checklists | Cards render |
| 10 | SENTINEL: fetch, hash, classify, targeted alert; injected demo circular | Alert flow |
| 11 | Trust Layer polish; agent trace panel; run evaluation; record numbers | Accuracy slide data |
| 12 | Deck, demo script, backup video | Deck v1, video |
| 13 | Two full dry runs with fresh people; fix top 5 issues | Stable demo |
| 14 | Buffer; rehearse Q and A; freeze code | Submission |

## If you only have 48 hours

- **Hours 0-4**: web chat UI, FastAPI skeleton, profile schema, 10 schemes with sources.
- **Hours 4-14**: voice → profile; GST certificate extraction with validators; calendar.
- **Hours 14-24**: rules engine, 3 scheme cards, Health Score; consent screen and disclaimer.
- **Hours 24-34**: injected SENTINEL alert; agent trace; PATHWAY checklist for one flow.
- **Hours 34-42**: run 15 documents for a measured number; fix errors; record backup video.
- **Hours 42-48**: deck, two dry runs, freeze.

## Team split (4 people; merge roles if fewer)

| Role | Owns | Hands off to |
|---|---|---|
| A: Agents and backend lead | ARIA, VEDA, orchestrator, validators, trace | B (API contract), C (profile schema) |
| B: Channel and UI | WhatsApp or web chat, cards, calendar view, demo polish | D (screens for deck) |
| C: Data and rules | Scheme DB, penalty rules, SCOUT engine, SENTINEL sources, evaluation set | A (rules JSON), D (numbers) |
| D: Research and pitch | Interviews, legal calls, fact-check, deck, Q and A, demo script | All (feedback loop) |

## Evaluation plan (this produces your headline numbers)

| What | Test set | Metric to report |
|---|---|---|
| Document extraction | 30 documents: 10 GST certificates, 10 notices or challans, 10 invoices or licences; include 5 photos taken on a phone, 5 in Hindi | Field-level exact match; share of wrong fields that were flagged low-confidence |
| Speech-to-text | 20 Hindi and Hinglish clips from 5 speakers, including noisy shop audio | Word error rate; profile fields captured correctly |
| Scheme matching | 10 synthetic profiles with hand-labelled eligible schemes | Precision and recall; 'not sure' rate |
| SENTINEL | 10 real official notices | Relevance precision; false-alert count |
| Usability | 5 people who have not seen it | Time to first useful answer; trust rating 1 to 5 |

**Data hygiene**
Use synthetic or consenting-owner documents for the demo and mask PAN and GSTIN on slides. Never put a real stranger's identifiers in a public deck.

## Starter code 1: validators and penalty rules (tested)

Rates and caps are from Section 2 and must be re-checked on the day you ship. GSTIN checksum logic was run against a known-valid sample before inclusion.

```
import re
from datetime import date

CH = "0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZ"
GSTIN_RE = re.compile(r"^[0-9]{2}[A-Z]{5}[0-9]{4}[A-Z][1-9A-Z]Z[0-9A-Z]$")
PAN_RE = re.compile(r"^[A-Z]{5}[0-9]{4}[A-Z]$")

def gstin_valid(g: str) -> bool:
    """Format + checksum. Catches most OCR misreads before anything is saved."""
    g = g.strip().upper()
    if not GSTIN_RE.match(g):
        return False
    s = 0
    for i, ch in enumerate(g[:14]):
        v = CH.index(ch) * (1 if i % 2 == 0 else 2)
        s += v // 36 + v % 36
    return CH[(36 - s % 36) % 36] == g[14]

def pan_valid(p: str) -> bool:
    return bool(PAN_RE.match(p.strip().upper()))

def gst_late_fee(days_late: int, nil_return: bool, turnover_cr: float) -> int:
    """Section 47 late fee for GSTR-1 / GSTR-3B (rates and caps: verify on the day you ship)."""
    per_day = 20 if nil_return else 50
    if nil_return:
        cap = 500
    elif turnover_cr <= 1.5:
        cap = 2000
    elif turnover_cr <= 5:
        cap = 5000
    else:
        cap = 10000
    return min(max(days_late, 0) * per_day, cap)

def gst_interest(tax_paid_late: float, days_late: int) -> float:
    """Section 50 interest, 18% per annum."""
    return round(tax_paid_late * 0.18 * max(days_late, 0) / 365, 2)

def days_late(due: date, today: date) -> int:
    return max((today - due).days, 0)
```

## Starter code 2: scheme eligibility engine (tested)

Unknown inputs return 'possible', not 'eligible'. The criteria values for any real scheme must be copied from the official notification and re-checked.

```
SCHEME = {
  "id": "cm_yuva_up",
  "name": "CM YUVA (Uttar Pradesh)",
  "source_url": "https://msme.up.gov.in",       # official page; store retrieved_on + checked_by
  "max_benefit_inr": 500000,                    # project-cost cap, NOT a promised payout
  "criteria": [
    {"field": "state",        "op": "eq",      "value": "UP",      "hard": True},
    {"field": "owner_age",    "op": "between", "value": [21, 40],  "hard": True},
    {"field": "has_skill_cert","op": "eq",     "value": True,      "hard": False},
    {"field": "has_other_subsidy","op": "eq",  "value": False,     "hard": True},
  ],
}

def _ok(op, have, want):
    if have is None:
        return None                      # unknown -> ask the user, do not guess
    if op == "eq":
        return have == want
    if op == "between":
        return want[0] <= have <= want[1]
    if op == "in":
        return have in want
    if op == "gte":
        return have >= want
    raise ValueError(op)

def evaluate(profile: dict, scheme: dict) -> dict:
    met, failed, unknown = [], [], []
    for c in scheme["criteria"]:
        r = _ok(c["op"], profile.get(c["field"]), c["value"])
        (unknown if r is None else met if r else failed).append(c)
    hard_fail = any(c["hard"] for c in failed)
    total = len(scheme["criteria"])
    return {
        "scheme": scheme["id"],
        "eligible": (not hard_fail) and not unknown,
        "possible": (not hard_fail) and bool(unknown),
        "score": f"{len(met)} of {total} criteria met",
        "missing_info": [c["field"] for c in unknown],
        "source_url": scheme["source_url"],
    }
```

## Starter prompt 1: VEDA document extraction

```
SYSTEM: You extract fields from an Indian business document image. Return ONLY JSON.
Never guess. If a field is unreadable or absent, set value to null and confidence to 0.
Schema:
{
  "doc_type": "gst_certificate | gst_notice | invoice | licence | challan | other",
  "fields": [
    {"name": "gstin|pan|legal_name|trade_name|issue_date|due_date|expiry_date|amount_inr|section_cited|notice_ref",
     "value": "string or null",
     "confidence": 0.0-1.0,
     "evidence": "exact text snippet you read it from"}
  ],
  "language_detected": "hi|en|mixed|other",
  "needs_human": true|false,
  "needs_human_reason": "string or null"
}
Rules: dates as YYYY-MM-DD. Amounts as digits only. Quote evidence verbatim.
POST-PROCESS (code, not LLM): validate GSTIN checksum, PAN regex, date plausibility;
if any check fails or confidence < 0.80 -> show field as "please confirm", never auto-save.
```

## Starter prompt 2: ARIA onboarding

```
SYSTEM: You are ARIA, a Hindi-first onboarding assistant. Speak simple Hindi or Hinglish, short sentences.
Goal: fill this profile. Ask ONE missing field at a time, most valuable first. Explain WHY you ask.
Never ask for passwords, OTPs or full Aadhaar. Return JSON: {"reply": "...", "profile_patch": {...}, "next_field": "..."}
PROFILE: {"business_type":null,"sector":null,"state":null,"district":null,"turnover_band":null,
 "employees":null,"gst_registered":null,"udyam_registered":null,"owner_gender":null,"owner_age":null,
 "has_skill_cert":null,"has_other_subsidy":null,"language":"hi"}
```

## Suggested repository layout

```
complianceos/
  app/
    main.py              # FastAPI: /whatsapp webhook, /web chat, /ca read-only
    orchestrator.py      # routes messages to agents, writes trace events
    agents/aria.py  veda.py  scout.py  sentinel.py  pathway.py
    trust/validators.py  confidence.py  consent.py  audit.py
    rules/rules.py       # Starter code 1
    rules/eligibility.py # Starter code 2
    data/schemes/*.json  # one file per scheme: criteria + source_url + retrieved_on + checked_by
    data/reg_feed/       # fetched notices (raw text + url + hash)
  eval/
    docs/ labels.csv     # 30 documents + ground truth
    run_eval.py          # prints the numbers for the deck
  web/                   # PWA fallback + agent trace panel
  README.md              # what is real, what is simulated
```

## 3-minute demo script

| Time | On screen | Say |
|---|---|---|
| 0:00 | Owner photo or persona card | One real sentence from an interview about a missed deadline |
| 0:20 | Hindi voice note → profile builds live | 'She just talks. ARIA builds her business profile and asks only what is missing.' |
| 0:50 | Photo of GST notice → extracted fields with confidence → 'please confirm' on one field | 'Low confidence is never saved silently. She confirms in one tap.' |
| 1:20 | Calendar fills; Health Score; penalty at risk with section cited | 'The number comes from code and law, not from a guess.' |
| 1:45 | Three scheme cards: 'X of Y met', source link | 'Each match shows why, what is missing, and the official page.' |
| 2:15 | Injected circular → targeted alert (labelled simulated) + agent trace | 'Only affected owners get this. Here is every agent call.' |
| 2:40 | 'Send to my CA' + trust panel | 'We guide; the CA approves. We never log in or file for her.' |
| 2:55 | Measured accuracy slide | '87%' style number only if you measured it |

# 13. Pitch deck: slide-by-slide content

Ten slides, one idea each, one big number per slide. Spend half the time on the demo. Speaker notes are written to be spoken in about 20 to 30 seconds.

| # | Title (the claim) | On-slide content | Visual | Speaker note |
|---|---|---|---|---|
| 1 | Nobody tells small owners what they owe | One real quote or story from your interviews; name, trade, city; ₹ cost of the miss. Tagline: 'Your compliance co-pilot, in your language, on WhatsApp' | Photo or persona card | Start with the person, not the market. Pause after the rupee amount. |
| 2 | 8.7 crore businesses, almost all micro, mostly unguided | 8.7 cr registrations [S1]; 99% micro [S4]; only 19% of credit demand met [S5]; up to ₹48,000 a year GST late fees alone | 3 big numbers + small footnotes with sources | Cite sources aloud once. Say 'worst case' for the ₹48,000. |
| 3 | We start from awareness, not filing | Pipeline: ARIA → VEDA → SCOUT → SENTINEL → PATHWAY around one Business Profile | One clean diagram | Filing tools help people who know what to file. We help people who do not know what they do not know. |
| 4 | Live demo | Run Section 12 script; show agent trace | Live product | Tell them what is simulated before they ask. |
| 5 | Why you can trust it | Citations, confidence gate, consent, CA in the loop, deterministic law; no login, no auto-filing | Screenshot of 'please confirm' card | This is where technical judges decide. Name the hallucination risk first. |
| 6 | Beachhead: Hindi-speaking GST traders and women-led units in UP | 1.67 cr GST registrations [S7]; UP leads new registrations; SAM and SOM with stated assumptions | Funnel graphic | Show assumptions as assumptions. Judges trust that. |
| 7 | Where we win | Matrix: ClearTax, IRIS Peridot, VendorWise, portals vs voice, WhatsApp, documents, schemes + deadlines, CA layer | Comparison table with ticks | Acknowledge IRIS Peridot on schemes. Win on voice, documents and trust. |
| 8 | It pays for itself | ₹299 plan; cost per paid user and margin from Section 9; capped free tier; CA flat subscription | Bar: revenue vs cost | Say the margin number and one assumption you still need to measure. |
| 9 | Go-to-market and 12-month path | CA partners, trade associations, DICs, CSCs; milestones; DPDP dates [S19] | Timeline | We do not buy ads. We borrow trust from CAs and associations. |
| 10 | Team and ask | Team strengths; interviews done; pilot plan; the specific ask (mentorship, pilot access, credits) | Team photo + 3 results | End with one result: a measured number or a pilot commitment. |

## Design rules for the deck

- One idea per slide; at most 25 words of body text; numbers at 40pt or larger.
- Reuse the palette (orange, green, gold, purple) from your current deck, but simplify layouts.
- Delete the SWOT slide; replace with slides 6 and 7. Delete the '5 agents wheel' slide; replace with the pipeline diagram.
- Fix known defects in the old deck: overlapping SENTINEL text and stray lines on the solution slide; empty card areas on the problem slide; faded 'Bank Referral' row; inconsistent price ranges.
- Footnote every statistic with its [S#] source in small type. Prepare a backup slide with the sources list.
- Record a 3-minute backup demo video; keep it on the laptop and a phone.

# 14. Tough questions and strong answers

| Question | Answer |
|---|---|
| What if the AI gives a wrong deadline? | Penalties and eligibility come from code and a verified database, not model memory. Every claim shows a source and last-verified date. Low-confidence fields need user confirmation, and high-stakes items go to a CA. |
| Why can't ClearTax or a fintech just add this? | They can, so we do not claim a permanent moat. They serve users who already know what to file, in English. Our wedge is vernacular voice, awareness first, a verified scheme database, and CA relationships built through partners. We will measure retention to prove it. |
| What about IRIS Peridot's scheme discovery? | It is the closest overlap. It is a GST-suvidha-provider app. We differ on WhatsApp-first voice, document-to-calendar, and a visible trust layer tied to a CA workflow. Our pilot will test whether owners prefer that. |
| How do you make money, and does it pay? | ₹299 plan and a flat CA subscription. Variable cost per paying user is about ₹60 on stated assumptions, so margin is near 80%. The free tier is capped because uncapped free users erode margin. |
| Is taking a CA commission legal? | We dropped it. ICAI rules limit solicitation and fee sharing, so we charge CAs a flat subscription and take nothing from their fees. We are confirming this with a practising CA. |
| Do you log into government portals? | No. We guide with checklists and official deep links. We never store credentials, OTPs or file on a user's behalf. |
| How do you handle privacy? | Explicit opt-in on first message, purpose-limited data, India-region encrypted storage, one-tap delete, and a plan aligned with the DPDP Rules, with full obligations from 13 May 2027 [S19]. |
| How accurate is your document reading? | State only what you measured: 'X% field-level exact match on 30 documents; Y% of errors were flagged low-confidence.' If you have not measured, say the test is running. |
| Why UP and why Hindi first? | UP leads new GST registrations [S7], has rich schemes (ODOP, CM YUVA, MSME Policy 2022 [S23]) and one language covers the pilot. Expansion is a per-language STT and content task. |
| Low digital literacy will slow adoption. | Voice notes and WhatsApp are the interface people already use. We also route through CAs, CSCs and trade bodies for assisted onboarding. |
| What if government releases its own tool? | Likely partial. We would integrate as a partner layer in local language and CA workflow; our verified rules database and relationships remain useful. |
| What do you need that you do not have? | Name it honestly: pilot access, a CA mentor, API credits, a legal review. Honest asks score well. |

# 15. Self-assessment: score and what moves it

| Criterion (weight) | Original | After full build + pilot data | What moves it |
|---|---|---|---|
| Problem and relevance (15) | 13 | 14 | Real quote + sourced numbers |
| Innovation (15) | 9 | 12 | Voice + documents + trust layer as one flow; sharper differentiation |
| Feasibility (20) | 10 | 16 | Deterministic rules, no portal automation, measured accuracy |
| Business model (15) | 9 | 12 | Three lines, unit economics, ICAI-safe, capped free tier |
| Social impact (10) | 9 | 9 | Vernacular, women-led, Tier 2/3 reach; keep it, do not oversell |
| Demo and execution (15) | 6 | 13 | Working vertical slice, agent trace, backup video |
| Pitch clarity (10) | 6 | 8 | One story, one number per slide, honest labels |
| **Total (100)** | **62** | **84** | Projection if the plan is executed; not a guarantee |

Where this can still fail: a demo that breaks live, an unmeasured accuracy claim, an unsourced statistic, or an overbroad product story. All four are in your control.

# 16. Risk register

| Risk | Likelihood | Impact | Mitigation |
|---|---|---|---|
| Live demo fails (network, API, WhatsApp approval) | Medium | High | Web fallback, recorded backup video, offline sample inputs |
| OCR errors on handwritten or regional documents | High | Medium | Confidence gate; present results honestly; scope to printed documents first |
| Wrong rate or deadline in rules DB | Medium | High | Source, date and checker on every rule; re-verify before demo; show 'last verified' |
| Legal objection (ICAI, GSP/ASP, DPDP) | Medium | High | Flat CA fee, no login or filing, consent flows; get one CA/lawyer opinion |
| Competitor overlap (IRIS Peridot, ClearTax) | Medium | Medium | Differentiate on voice, documents, trust layer; show comparison |
| Cold start: thin CA network | High | Medium | Start with 3 to 5 friendly CAs, not a marketplace |
| Cost blowout from free users | Medium | Medium | Cap free tier; track per-user cost from day one |
| Rule changes (WhatsApp pricing, GST, labour) | High | Low to medium | Quarterly re-check; keep rates in config, not code |
| Scope creep in the last week | High | High | Freeze scope at day 10; P2 items only if P0 and P1 are stable |

# 17. Verification checklist and sources

## Not verified in our research: check before presenting

- The legal position on third-party GST portal access (GSP/ASP) for any future automation.
- RBI digital-lending and DSA rules, if you ever add loan referral.
- Zoho, Tally, Vyapar, Vakilsearch and Khatabook compliance features; confirm before claiming a gap.
- Number of CAs and tax practitioners in India, and typical CA fees; run your own calls.
- State-wise active GST counts for your beachhead; use the Department of Revenue dataset.
- Which states have final Labour Code rules today; trackers disagree on counts, so check the state gazette.
- WhatsApp service-message charges after 1 Oct 2026; sources differ, so read Meta's own rate card.
- Every scheme's current eligibility, caps and portal status, from the official notification.

## Sources

| Tag | Source and what we used it for | Link |
|---|---|---|
| S1 | Ministry of MSME via UNI, 27 Jun 2026: GDP 31.1%, manufacturing 35.4%, exports 48.58%, 8.7 crore Udyam + Udyam Assist registrations. | https://www.uniindia.com/~/india-s-msme-sector-contributes-31-1pc-to-gdp-formal-registrations-cross-8-7-cr/Business%20Economy/news/3891190.html |
| S2 | PIB, Mar 2026: 7.83 crore Udyam registrations till 28 Feb 2026. | https://www.pib.gov.in/PressReleasePage.aspx?PRID=2246892®=3=1 |
| S3 | PIB, Jul 2025: 30.1% GDP; Udyam Assist 2.72 crore informal micro units. | https://www.pib.gov.in/PressReleasePage.aspx?PRID=2142170®=48=2 |
| S4 | Policy Circle, Apr 2025: 6.11 of 6.19 crore registered MSMEs are micro; credit penetration 14%. | https://www.policycircle.org/policy/india-msmes-funding-gap/ |
| S5 | NITI Aayog / IFC report via PIB: 19% of credit demand formally met (FY21), ₹80 lakh crore unmet; low awareness limits policy effectiveness. | https://www.pib.gov.in/PressReleasePage.aspx?PRID=2126063®=48=2 |
| S6 | Deloitte via Whalesbook, Jun 2026: about ₹25 lakh crore MSME credit gap; 14% formal access. | https://www.whalesbook.com/news/English/economy/Indias-Rs-25-Trillion-MSME-Credit-Gap-What-It-Means-For-Lenders/6a3fa66afb97bcde4ce4fa35 |
| S7 | GSTN data via A2Z Taxcorp / BusinessLine, Jul 2026: 1.67 crore registrations at 30 Jun 2026; UP leads new registrations; e-invoicing threshold ₹5 cr from Apr 2026. | https://a2ztaxcorp.net/nearly-8-lakh-gst-registrations-in-6-months-up-leads-the-way/ |
| S8 | GST Day 2026 summary: GST 2.0 effective 22 Sep 2025. | https://www.studyiq.com/articles/gst-day-2026/ |
| S9 | GST late fee and interest guides (IncorpX, TaxGarden, ClearTax): s.47 rates and caps; s.50 interest 18%. | https://www.incorpx.io/blog/gst-late-fees-penalties-rate-chart-2026 \| https://taxgarden.in/blog/gst-late-fee-interest-penalty-gstr-3b-gstr-1-gstr-9-guide \| https://cleartax.in/s/gst-return-late-fees |
| S10 | Finance Ministry data via Deccan Chronicle, Dec 2023: 90% of eligible taxpayers file GSTR-3B by month-end (68% in 2017-18). | https://www.pressreader.com:443/india/deccan-chronicle/20231218/282033331992689 |
| S11 | DLA Piper, May 2026: central rules under all four Labour Codes notified 8 May 2026. | https://knowledge.dlapiper.com/dlapiperknowledge/globalemploymentlatestdevelopments/2026/Key-considerations-of-the-notified-Central-Rules-under-Indias-Labour-Codes |
| S12 | KPMG flash alert, May 2026: final rules under four Labour Codes. | https://kpmg.com/xx/en/our-insights/gms-flash-alert/2026/flash-alert-2026-127.html |
| S13 | LKS Law labour code rules tracker: central and state status. | https://employmentlaw.lkslaw.com/rules |
| S14 | Beacon Filing tracker: states with final rules for all four codes as of Mar 2026 (counts differ across trackers). | https://beaconfiling.com/blog/labour-code-implementation-tracker-states |
| S15 | SIDBI: GST Sahay invoice-financing application (OCEN, Account Aggregator). | https://www.sidbi.in/gst-sahay |

| Tag | Source and what we used it for | Link |
|---|---|---|
| S16 | IRIS Peridot app listing (GSP/IRP-backed; GST checks, MSME scheme discovery). ClearTax GST automation context: cleartax.in/s/gstr-3b. | https://apps.apple.com/us/app/iris-peridot/id1406656314?ls=1 |
| S17 | VendorWise AI: compliance document and expiry tracking for Indian SMEs. | https://vendorwiseai.com/ |
| S18 | ICAI Code of Ethics: BCAJ on clause 6; ICAI FAQ on clauses 2, 6, 7; Technocratiq on 13th Edition (from 1 Apr 2026). | https://bcajonline.org/journal/code-of-ethics-4/ \| https://www.icagaurav.com/2025/03/mastering-ethical-standards-faqs-for.html \| https://technocratiq.com/insights/digital-marketing-for-chartered-accountants/ |
| S19 | DPDP Rules 2025 timeline and penalties (Sansa Legal; ComplianceHub; PIB explainer). | https://www.sansalegal.com/post/dpdp-act-2023-and-rules-2025-phased-implementation-timeline-and-business-compliance-deadlines \| https://compliancehub.wiki/india-dpdp-consent-manager-november-2026-phase-two-deadline-compliance/ |
| S20 | WhatsApp Business Platform pricing: Meta developer docs; INR rate card summaries. | https://developers.facebook.com/documentation/business-messaging/whatsapp/pricing \| https://chatmaxima.com/whatsapp-api-pricing/india/ \| https://whautomate.com/whatsapp-business-api-pricing-india |
| S21 | Sarvam speech-to-text (22 languages, code-mixing, ₹1.5/min) and Sarvam Startup Program. | https://www.sarvam.ai/speech-to-text \| https://en.wikipedia.org/wiki/Sarvam_AI |
| S22 | Bhashini APIs documentation; open-source Indian voice AI overview. | https://bhashini.gitbook.io/bhashini-apis \| https://caller.digital/blog/open-source-voice-ai-india-sarvam-ai4bharat-bhasini-2026 |
| S23 | UP MSME schemes: ODOP, UP MSME Policy 2022, CM YUVA (TaxGuru; IIFL; Utkarsh SFB). | https://taxguru.in/corporate-law/uttar-pradesh-msme-subsidy-schemes-guide-grants-loans-incentives.html \| https://www.iifl.com/blogs/business-loan/up-msme-subsidy-schemes \| https://www.utkarsh.bank.in/business/loans/mukhyamantri-yuva-udyami-vikas-yojana |

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